

BASICS OF WEB

➤ **Network:**

- Network is 2 or more computers are connected to each other and can share data with each other.
- There are three types of networks –
 - ❖ LAN: Local Area Network. In this network generally, the different devices are connected with each other using wired network.
 - ❖ MAN: Network covers an entire town area or a portion of a city. Data transmission speed is relatively high due to the installation of optical cables and wired connections.
 - ❖ WAN: The Wide Area Network is designed to connect devices over large distances like states or between countries. The connection is wireless in most cases and uses radio towers or satellites for communication.

➤ **Internet:**

- The Internet is essentially a global network of computing resources.
- The Internet is a physical collection of routers and circuits as a set of shared resources.

➤ **WWW:**

- www is World Wide Web.
- A technical definition of the World Wide Web is – All the resources and users on the Internet that are using the Hypertext Transfer Protocol (HTTP).

➤ **W3C:**

- The World Wide Web Consortium (W3C) is an international community where Member organizations, full-time staff, and the public work together to develop Web standards.

➤ **Communication Protocols: Network Protocols** are a set of rules governing exchange of information between two or more entities in an easy, reliable and secure way.

- Transmission Control Protocol (TCP): TCP is a popular communication protocol which is used for communicating over a network. It **divides any message into series of packets that are sent from source to destination and there it gets reassembled at the destination.**
- Internet Protocol (IP): IP is designed explicitly as addressing protocol. It is mostly **used with TCP**. The **IP addresses in packets help in routing them through different nodes** in a network until it reaches the destination system. TCP/IP is the most popular protocol connecting the networks.
- User Datagram Protocol (UDP): UDP is **commonly used in time-sensitive communications where occasionally dropping packets is better than waiting.** Voice and video traffic are often sent using this protocol because they are both time-sensitive and designed to handle some level of loss.
- Post office Protocol (POP): POP3 is designed for **receiving incoming E-mails.**

- Simple mail transport Protocol (SMTP): SMTP is designed to **send and distribute outgoing E-Mail**.
- File Transfer Protocol (FTP): FTP allows users to **transfer files from one machine to another**. Types of files may include program files, multimedia files, text files, and documents, etc.
- Hyper Text Transfer Protocol (HTTP): HTTP is used **for transferring data between the client browser (request) and the web server (response) in the hypertext format**.
- Hyper Text Transfer Protocol Secure (HTTPS): HTTPS is abbreviated as Hyper Text Transfer Protocol Secure is a **standard protocol to secure the communication among two computers** one using the browser and other fetching data from web server. HTTPS except that the transferring of data is done in an encrypted format.
- Telnet: Telnet is a set of rules designed for connecting one system with another. The connecting process here is **termed as remote login**. The system which requests for connection is the local computer, and the system which accepts the connection is the remote computer.
- Gopher: Gopher is a collection of rules implemented for **searching, retrieving as well as displaying documents from isolated sites**. Gopher also works on the client/server principle.

➤ **Web Page:**

- A web page is a simple document displayable by a browser.
- Such documents are written in the HTML language (which we look into in more detail in other articles).

➤ **Web Site:**

- Collection of various pages written in HTML markup language is called a website.
- This is a location on the web where people can find information on the latest technologies.
- Each page available on the website is called a web page and the first page of any website is called the home page for that site.

➤ **Web Server:**

- A web server is a computer hosting one or more websites.
- "Hosting" means that all the web pages and their supporting files are available on that computer.
- The web server will send any web page from the website it is hosting to any user's browser, per user request.

➤ **Web Browser:**

- Web Browsers are software installed on your PC.
- To access the Web, you need web browsers, such as Microsoft Edge, Google Chrome or Mozilla Firefox.

- On the Web, when you navigate through pages of information this is commonly known as browsing or surfing.
- **ISP:**
 - ISP stands for Internet Service Provider. They are the companies who provide you service in terms of internet connection to connect to the internet.
 - You will buy space on a Web Server from any Internet Service Provider. This space will be used to host your Website.
- **DNS:**
 - DNS means Domain Name System.
 - **It translates human readable domain names (for example, www.amazon.com) to machine readable IP addresses (for example, 192.0.2.44).**
- **URL:**
 - URL stands for Uniform Resource Locator.
 - URL is used to specify addresses on the World Wide Web.
 - A URL is the fundamental network identification for any resource connected to the web (e.g., hypertext pages, images, and sound files).
- **Query String:**
 - A query string is a **set of characters** tacked onto the **end of a URL**.
 - The query string **begins** after the **question mark (?)** and can include one or more parameters.
 - Each **parameter is represented by a unique key-value pair** or a set of two linked data items.
 - An equals sign (=) **separates each key and value**.
 - An **ampersand (&)** **separates multiple parameters**.
- **Request-Response Architecture:**
 - Your personal computer: This is the PC at which you sit to see the web. A Web browser: Software installed on your PC which helps you to browse the Web.
 - An internet connection: This is provided by an ISP and connects you to the internet to reach any Website.
 - A Web server: This is the computer on which a website is hosted.
 - Routers & Switches: They are the combination of software and hardware that take your request and pass it to the appropriate Web server.
 - The Web is known as a client-server system. Your computer is the client and the remote computers that store electronic files are the servers.